# Algorithm: Train Coach Management Using Stack and Queue

1. Start.

2. Initialize an empty stack for attached coaches and an empty queue for waiting coaches.

3. Display the menu of operations.

4. Accept the user's choice.

5. If the choice is Add Coach, enter the coach number and type, then push the coach onto the stack.

6. If the choice is Remove Coach, check whether the stack is empty. If it is empty, display a message; otherwise, pop and remove the top coach.

7. If the choice is Add Waiting Coach, enter the coach number and type, then insert the coach into the queue using enqueue().

8. If the choice is Attach Waiting Coach, check whether the queue is empty. If it is empty, display a message; otherwise, dequeue the first waiting coach and push it onto the stack.

9. If the choice is Display Coaches, display all attached coaches in the stack and all waiting coaches in the queue.

10. If the choice is Exit, stop the program.

11. If the choice is invalid, display an invalid-choice message.

12. Repeat Steps 3–11 until the user selects Exit.

13. Stop.